MILITARY SPECIFICATION SHEET

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED, LOW-POWER SCHOTTKY TTL, FLIP-FLOPS, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/371A, dated 8 August 1986, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 28 and 29

* FIGURE 5, NOTE 2, t_p: Delete "15 ns" and substitute "17 ns".

PAGE 35

FIGURE 5, NOTE 2: Delete "PRR is 50% of the clock PRR".

PAGE 36

FIGURE 5, NOTE 3: Delete "5%" and substitute "1%."

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
5	5
6	6
33	Reprinted without change
34	34

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PAGE 49

* TABLE III, device type 01, test numbers 45 through 54, V_{CC} column: Delete "5.0 V" and substitute "5.5 V".

* TABLE III, device type 01, I_O, test numbers 49 through 52, Min and Max limits column: Delete "-30" and "-112" and substitute "10/".

* TABLE III, device type 01, I_{CC}, test numbers 53 and 54, maximum limits column: Delete "40" and substitute "4.0".

TABLE III, Subgroup 7, Symbol column: Add "9/".

* TABLE III, device type 01, truth table tests, test number 63, 1Q, $1\overline{Q}$, $2\overline{Q}$, and 2Q columns: Delete "„" and substitute "H", "L", "L", and "H", respectively.

* TABLE III, device type 01, truth table tests, test number 79, 1Q, $1\overline{Q}$, $2\overline{Q}$, and 2Q columns: Delete "L", "H", "H", and "L" and substitute "H", "L", "L", and "H", respectively.

* TABLE III, device type 01, truth table tests, test number 81, 1Q, $1\overline{Q}$, $2\overline{Q}$, and 2Q columns: Delete "H", "L", "L", and "H" and substitute "L", "H", "H", and "L", respectively.

PAGE 51

Add new footnote 9/ as follows:

"9/ A = 3.0 V minimum; B = 0.0 V or GND."

* TABLE III, add footnote 10/ as follows:

"

Test	Min/Max limits in mA for circuit		
	A	B	C
I_O	-20/-112	-30/-112	-30/-112

"

PAGE 53

* TABLE III, device type 02, I_O, test numbers 57 through 60, Min and Max limits column: Delete "-30" and "-112" and substitute "11/".

TABLE III, Subgroup 7, Symbol column: Add "10/".

PAGE 54

* TABLE III, device type 02, t_{PLH1}, test number 107, measured terminal column: Delete "1$\overline{CLR}$ to 1Q" and substitute "1CLR to 1Q".

* TABLE III, device type 02, t_{PLH1}, test number 108, 1$\overline{PR}$ column: Delete "3.5 V" and substitute "IN".

* TABLE III, device type 02, t_{PHL1}, test number 111, measured terminal column: Delete "1$\overline{CLR}$ to $1\overline{Q}$" and substitute "1CLR to 1Q".

* TABLE III, device type 02, t_{PHL1}, test number 112, measured terminal column: Delete "1$\overline{PR}$ to 1Q" and substitute "1PR to 1Q".

* TABLE III, device type 02, t_{PHL1}, test number 113, measured terminal column: Delete "2$\overline{PR}$ to 2Q" and substitute "2PR to 2Q".

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* TABLE III, device type 02, t_{PHL1}, test number 114, measured terminal column: Delete "2$\overline{CLR}$ to 2$\overline{Q}$" and substitute "2CLR to 2Q".

* TABLE III, device type 02, t_{PLH2}, test number 115, measured terminal column: Delete "1CLR to 1$\overline{Q}$" and substitute "1CLK to 1Q".

* TABLE III, device type 02, t_{PLH2}, test number 117, measured terminal column: Delete "2CLR to 2Q" and substitute "2CLK to 2Q".

* TABLE III, device type 02, t_{PLH2}, test number 118, measured terminal column: Delete "2CLK to 2$\overline{Q}$" and substitute "2CLK to 2Q".

* TABLE III, device type 02, t_{PHL2}, test number 121, measured terminal column: Delete "2CLK to 2Q" and substitute "2CLK to 2Q".

* TABLE III, device type 02, t_{PHL2}, test number 122, measured terminal column: Delete "2CLK to 2$\overline{Q}$" and substitute "2CLK to 2Q".

PAGE 55

Add new footnote 10/ as follows:

"10/ A = 3.0 V minimum; B = 0.0 V or GND."

* TABLE III, add footnote 11/ as follows:

"

Test	Min/Max limits in mA for circuit		
	A	B	C
I_O	-20/-112	-30/-112	-30/-112

"

PAGE 57

TABLE III, I_{IH5}, Max limits column: Delete "10" and substitute "20".

TABLE III, Subgroup 7, Symbol column: Add "10/".

PAGE 58

* TABLE III, device type 03, t_{PLH1}, test number 99, measured terminal column: Delete "1CLR to 1$\overline{Q}$" and substitute "1CLR to 1Q".

* TABLE III, device type 03, t_{PLH1}, test number 100, measured terminal column: Delete "2$\overline{CLR}$ to 2Q" and substitute "2CLR to 2Q".

* TABLE III, device type 03, t_{PLH1}, test number 101, measured terminal column: Delete "2$\overline{PR}$ to 2Q" and substitute "2PR to 2Q".

* TABLE III, device type 03, t_{PHL1}, test number 104, 2$\overline{Q}$ column: Delete "OUT".

* TABLE III, device type 03, t_{PHL1}, test number 104, 2Q column: Add "OUT".

* TABLE III, device type 03, t_{PHL1}, test number 104, measured terminal column: Delete "2$\overline{CLR}$ to 2$\overline{Q}$" and substitute "2CLR to 2Q".

* TABLE III, device type 03, t_{PHL1}, test number 105, 2Q column: Delete "OUT".

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* TABLE III, device type 03, t_{PHL1}, test number 105, $2\overline{Q}$ column: Add "OUT".

* TABLE III, device type 03, t_{PHL1}, test number 105, measured terminal column: Delete "$2\overline{PR}$ to 2Q" and substitute "2PR to 2Q".

* TABLE III, device type 03, t_{PLH2}, test number 108, measured terminal column: Delete "2CLK to 2Q" and substitute "2CLK to 2Q".

* TABLE III, device type 03, t_{PLH2}, test number 109, measured terminal column: Delete "2CLK to $2\overline{Q}$" and substitute "2CLK to 2Q".

* TABLE III, device type 03, t_{PHL2}, test number 112, measured terminal column: Delete "2CLK to 2Q" and substitute "2CLK to 2Q".

* TABLE III, device type 03, t_{PHL2}, test number 113, measured terminal column: Delete "2CLK to $2\overline{Q}$" and substitute "2CLK to 2Q".

PAGE 59

Add new footnote 10/ as follows:

"10/ A = 3.0 V minimum; B = 0.0 V or GND."

* TABLE III, footnote 9/, I_O, circuit A: Delete "-30/-112" and substitute "-20/-112".

PAGE 60

* TABLE III, device type 04, top of the page, delete "Terminal conditions (pins not desinated may be high 2.0 V, low 0.8 V, or open)" and substitute "Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)".

* TABLE III, device type 04, V_{IC}, test number 25, 8D column: Add "-18 mA".

PAGE 61

* TABLE III, device type 04, top of the page, delete "Terminal conditions (pins not desinated may be high 2.0 V, low 0.8 V, or open)" and substitute "Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)".

* TABLE III, device type 04, I_O, test numbers 57 through 64, Min and Max limits column: Delete "-15" and "-110" and substitute "8/" (alters changes made by amendment 1).

TABLE III, I_{CCH}, Max Limits column: Delete "17" and substitute "18".

TABLE III, I_{CCL}, Max Limits column: Delete "23" and substitute "27".

TABLE III, I_{CCZ}, Max Limits column: Delete "27" and substitute "28".

TABLE III, Subgroup 7, Symbol column: Add "7/".

TABLE III, tests 84-90, column 11: Delete "L" and "H" and substitute "B" and "A" respectively.

TABLE III, f_{MAX}, Min Limits column: Delete "30" and substitute "35".

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* TABLE III, device type 04, top of the page, delete "Terminal conditions (pins not desinated may be high 2.0 V, low 0.8 V, or open)" and substitute "Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)".

TABLE III, t_{PLH2}, Max Limits column: Delete "12" and substitute "14".

TABLE III, t_{PLZ}, Min and Max Limits column: Delete "3" and "13" and substitute "2" and "12" respectively.

TABLE III, t_{PHZ}, Max Limits column: Delete "8" and substitute "10".

PAGE 63

* TABLE III, device type 04, top of the page, delete "Terminal conditions (pins not desinated may be high 2.0 V, low 0.8 V, or open)" and substitute "Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)".

TABLE III, t_{PLH2}, Max Limits column: Delete "15" and substitute "18".

TABLE III, t_{PHL2}, Max Limits column: Delete "„" and substitute "17".

TABLE III, t_{PLZ}, Min and Max Limits column: Delete "3" and "15" and substitute "2" and "18" respectively.

TABLE III, t_{PHZ}, Max Limits column: Delete "10" and substitute "12".

Add new footnote 7/ as follows:

"7/ A = 3.0 V minimum; B = 0.0 V or GND."

* TABLE III, add footnote 8/ as follows:

"

Test	Min/Max limits in mA for circuit		
	A	B	C
I_O	-20/-112	-30/-112	-30/-112

"

PAGE 64

* TABLE III, device type 05, top of the page, delete "Terminal conditions (pins not desinated may be high 2.0 V, low 0.8 V, or open)" and substitute "Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)".

PAGE 65

* TABLE III, device type 05, top of the page, delete "Terminal conditions (pins not desinated may be high 2.0 V, low 0.8 V, or open)" and substitute "Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)".

TABLE III, Subgroup 7, Symbol column: Add "7/".

PAGE 66

* TABLE III, device type 05, top of the page, delete "Terminal conditions (pins not desinated may be high 2.0 V, low 0.8 V, or open)" and substitute "Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)".

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PAGE 67

★ TABLE III, device type 05, top of the page, delete "Terminal conditions (pins not desinated may be high 2.0 V, low 0.8 V, or open)" and substitute "Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)".

Add new footnote 7/ as follows:

"7/ A = 3.0 V minimum; B = 0.0 V or GND."

PAGE 70

TABLE III, Subgroup 7, Symbol column: Add "8/".

PAGE 72

Add new footnote 8/ as follows:

"8/ A = 3.0 V minimum; B = 0.0 V or GND."

PAGE 75

TABLE III, Subgroup 7, Symbol column: Add "8/".

PAGE 77

Add new footnote 8/ as follows:

"8/ A = 3.0 V minimum; B = 0.0 V or GND."

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The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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